MSMT-LCL: Multiscale Spatial–Spectral Masked Transformer With Local Contrastive Learning for Hyperspectral Image Classification

Yunfei Zhou, Xiaohui Huang, *Member, IEEE*, Xiaofei Yang, *Member, IEEE*, Jiangtao Peng, *Senior Member, IEEE*, Yifang Ban, *Senior Member, IEEE*, and Nan Jiang, *Member, IEEE*

***Abstract*— Deep learning plays a crucial role in hyperspectral image (HSI) classification, with the Transformer being highly favored by researchers due to its exceptional ability to model long-range dependencies. However, the Transformer necessitates a substantial amount of labeled training samples to train its numerous parameters, exacerbating the challenge of training an effective HSI classification Transformer model, particularly given the inherent scarcity of HSI data. Therefore, we propose a novel method for HSI classification, termed multiscale spatial–spectral masked Transformer with local contrastive learning (MSMT-LCL). This method consists of two stages: self-supervised pretraining and supervised fine-tuning. Initially, we utilize the multiscale augmented feature mapping module (MAFM) to project original HSI data into two mixed-scale feature maps, which are then separately fed into two masked Transformer branches for reconstruction. To facilitate the model in learning the dependency relationships between central pixel land-cover information and neighboring land cover, we introduce a novel mask strategy based on center-patch. Furthermore, in the pretraining stage, we integrate local contrastive learning (LCL) to enable the model to focus on local center information at varying scales. Upon completion of pretraining, the network undergoes fine-tuning to obtain feature maps at two different scales. Subsequently, we devise a novel adaptive multiscale feature fusion module (AMFM) to adaptively aggregate these two features and produce the final classification results. Extensive experiments on three real datasets demonstrate the superiority of our proposed MSMT-LCL method over several state-of-the-art HSI classification methods.**

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Yunfei Zhou, Xiaohui Huang, and Nan Jiang are with the School of Information and Software Engineering, East China Jiaotong University, Nanchang 330013, China (e-mail: 2022068085404024@ecjtu.edu.cn; hxh016@hotmail.com; jiangnan1018@gmail.com).

Xiaofei Yang is with the School of Electronic and Communication Engineering, Guangzhou University, Guangzhou 511370, China (e-mail: xiaofeiyang@gzhu.edu.cn).

Jiangtao Peng is with Hubei Key Laboratory of Applied Mathematics, the Key Laboratory of Intelligent Sensing System and Security, Ministry of Education, and the Faculty of Mathematics and Statistics, Hubei University, Wuhan 430062, China (e-mail: pengjt1982@hubu.edu.cn).

Yifang Ban is with the Division of Geoinformatics, School of Architecture and the Built Environment, KTH Royal Institute of Technology, 114 28 Stockholm, Sweden (e-mail: yifang@kth.se).

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***Index Terms*— Contrastive learning (CL), deep learning, hyperspectral image (HSI) classification, masked Transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) typically consist of dozens to hundreds of spectral bands, providing rich spectral information in addition to spatial details [1], [2]. Compared to the three-channel representation of RGB images, this extensive spectral data enhances the ability to identify material characteristics [3], [4]. HSI classification is a critical task in the field of remote sensing image processing, as it involves assigning a true land-cover label to each pixel in an HSI. This technique has wide-ranging applications in agriculture [5], environmental monitoring [6], and aerospace industries [7].

Due to its excellent capability for deep feature extraction, deep learning has been widely applied in HSI classification tasks [8]. Mou et al. [9] constructed a network using an improved gated recurrent unit and proposed a novel activation function for analyzing HSI sequential data. In addition, Hang et al. [10] introduced a cascaded RNN model with gated recursion to improve existing RNN-based HSI classification methods.

Due to the excellent performance of convolutional neural networks (CNNs) in spatial feature extraction, many CNN-based methods have been employed for HSI classification [11]. Song et al. [12] proposed a deep feature fusion network to improve classification accuracy by integrating spatial features from different layers. Considering the 3-D nature of HSI images, Zhong et al. [13] designed an end-to-end 3-D convolutional residual network to extract fused spatial–spectral information. To combine the advantages of 2-D and 3-D convolutions, Yu et al. [14] developed a novel HSI classification method based on a simplified 2-D–3-D CNN through the collaboration of 2-D and 3-D convolutional layers.

In recent years, Transformer networks [15] have demonstrated exceptional performance in natural language processing due to their powerful long-range dependency modeling capabilities. Dosovitskiy et al. [16] introduced Transformers to the field of visual image processing, proposing the Vision Transformer (ViT) as a baseline model for visual tasks.

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In addition, many researchers have applied Transformers to HSI classification tasks. For example, He et al. [17] proposed a bidirectional encoder representation Transformer to capture global dependencies between pixels. Hong et al. [18] designed a Transformer from a sequential perspective based on grouped spectral embeddings, named SpectralFormer. Although these Transformer-based methods effectively consider long-range dependency modeling for spectral information, their modeling of local spatial features remains insufficient, resulting in suboptimal classification performance.

To address this issue, many researchers have combined the local modeling advantages of CNNs with the long-range modeling strengths of Transformers for spatial–spectral information extraction. Sun et al. [19] proposed a novel Transformer method for HSI classification, named SSFTT. This method incorporates a feature mapping module combining 2-D and 3-D convolutions to extract shallow spectral and spatial features, which are then transformed using a Gaussian-weighted feature tokenizer and input into the Transformer for feature extraction. In addition, Zhou et al. [20] proposed a visual Transformer using mobile 3-D convolutions as feature mappers and an inverted residual structure to reduce the number of model parameters.

Furthermore, some researchers have integrated CNNs into attention mechanisms to further blend the advantages of CNNs and Transformers. For instance, Zhang et al. [21] introduced convolution operations into the multihead attention mechanism and applied a group parallel residual block to capture local spectral–spatial features in HSI patches. Zhou et al. [22] proposed a novel convolutional Transformer architecture for HSI classification that models the height, width, and spectral dimensions, introducing a convolution-based interactive attention mechanism and designing parallel CNN and Transformer branches to extract spatial–spectral features.

Recently, He et al. [23] introduced a masked Transformer model for visual tasks, known as MAE. This model employs unsupervised pretraining with unlabeled samples followed by supervised fine-tuning with labeled samples, significantly reducing dependence on labeled data and enhancing model performance. Many researchers have adopted this pretraining and fine-tuning strategy for HSI classification tasks to mitigate the reliance on labeled samples. For instance, Ibañez et al. [24] utilized a grouped spectral encoding embedded in a random masked Transformer for unsupervised reconstruction self-learning, followed by supervised fine-tuning. In addition, Yang et al. [25] proposed a novel pretraining task that involves multiscale transformations and spectral sequence reversals on input HSI data, using a CNN to predict the scale and spectral sequence of the generated HSI.

Although the aforementioned methods have their own advantages, they still have some limitations that hinder further improvements in model accuracy. These limitations are summarized as follows.

1) For RNN-based methods [9], [10], although they demonstrate certain advantages in modeling spectral information, these RNN-based approaches neglect the importance of spatial data modeling, leading to suboptimal classification results.
2) For CNN-based methods [12], [13], [14], due to the good performance of CNNs in spatial feature extraction, these CNN-based methods significantly improve the ability to model spatial features. However, the modeling of spectral information remains insufficient, resulting in suboptimal utilization of information and unsatisfactory classification results.
3) For methods combining the CNN and Transformer [19], [20], while they partially address the traditional Transformer's weakness in extracting local spatial features, they still require a large number of labeled training samples for model training. However, since the acquisition of labeled HSI data is costly, these methods are limited in practical applications.
4) For methods based on the pretraining and fine-tuning strategy [24], [25], although these methods reduce reliance on labeled data to some extent, they overlook the importance of focusing on central spatial–spectral information during the modeling process, and the modeling of multiscale information is still insufficient, leading to a lack of further improvement in classification accuracy.

To address the aforementioned challenges, we propose a novel multiscale spatial–spectral masked Transformer architecture with local contrastive learning, termed multiscale spatial–spectral masked Transformer with local contrastive learning (MSMT-LCL). This architecture comprises a self-supervised pretraining stage and a supervised fine-tuning stage, effectively reducing reliance on labeled data. During the pretraining stage, we designed a multiscale augmented feature mapping module (MAFM) to obtain mixed-scale feature maps. Subsequently, a dual-branch masked Transformer is employed for self-supervised learning to reconstruct features at various scales, enabling the network to learn both local and global spatial–spectral information across different scales. Moreover, we have introduced a novel center-patch masking strategy to enable the model to better learn central spatial–spectral features. In addition, we incorporated local CL in the pretraining stage to focus the network on feature information near the center during reconstruction. In the fine-tuning stage, we developed a novel adaptive multiscale feature fusion module (AMFM) that adaptively merges the different multiscale features extracted by the model. This allows the model to more effectively capture and integrate spatial–spectral information at various scales. Our contributions are summarized as follows.

1) We propose a novel multiscale spatial–spectral masked Transformer method with local CL for HSI classification. MSMT-LCL uses a multiscale center-patch masking reconstruction with local contrastive learning (LCL), allowing the model to learn richer multiscale spatial–spectral feature representations from limited training samples.
2) We designed a novel center-patch masking strategy and an LCL strategy for the reconstruction task. These strategies enhance the model's ability to learn central multiscale spatial–spectral features during the self-supervised pretraining stage, thereby improving both reconstruction performance and key feature capture.

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3) We introduce an MAFM and an AMFM. An MAFM is used during the pretraining stage to obtain mixed-scale feature maps, while an AMFM is used to fuse different scale-mixed feature maps adaptively during the fine-tuning stage, enabling the model to more effectively capture and integrate multiscale spatial–spectral information, enhancing feature richness and classification accuracy.
4) We compared the proposed MSMT-LCL method with several state-of-the-art classification methods on three real datasets. Experimental results show that our method achieves the best performance with limited training samples.

The remainder of this article is structured as follows. Section II reviews related work in HSI classification. Section III provides a detailed description of the proposed MSMT-CLC method. In Section IV, we present and analyze the experimental results on three real-world datasets. Finally, Section V summarizes our work.

II. Related Work

In HSI classification tasks, obtaining accurately labeled HSIs is challenging. Consequently, many researchers are exploring self-supervised learning methods to achieve higher model performance with fewer labeled samples. In this section, we will discuss research on masked reconstruction-based generation methods, augmented view-based contrastive methods, and cascaded methods that integrate reconstruction generation and contrastive learning.

A. Masked Reconstruction-Based Generation Methods

Although deep learning methods exhibit unique advantages in HSI classification, they often require a large number of labeled training samples. To address this issue, researchers have proposed several notable methods [26], [27], [28], [29], [30]. For example, Scheibenreif et al. [26] introduced a self-supervised Transformer model based on masked image reconstruction. This model enhances HSI classification accuracy by optimizing block patch embedding, spatial–spectral self-attention, and spectral positional embedding, even with limited labeled samples. Liu et al. [27] developed a self-supervised learning method based on spectral masking (SSLSM), which masks and reconstructs spectral bands of input images, fine-tuning a cascaded encoder–decoder network for deep semantic feature extraction. In addition, Chen and Yan [28] proposed a novel self-supervised learning framework based on a low-frequency spectral masking strategy (LFSMIM) for HSI classification. LFSMIM reduces the network's sensitivity to noise by filtering out high-frequency components in the reconstruction target, effectively capturing detailed textures while leveraging global spectral features. Zhou et al. [31] designed a novel pretraining task for HSI classification, employing spatial and spectral mask branches to reconstruct local binary pattern features and spectral principal components. This approach enables the network to comprehensively learn spatial and spectral features of HSI during training.

Although these masked reconstruction-based generation methods have achieved some progress in HSI classification, they still have certain limitations. Most of them rely solely on the mask reconstruction task for feature learning, without fully leveraging the relationships between samples. This may prevent the model from capturing the complex structures and patterns in HSIs. Moreover, the masking strategies employed may not fully harness the key information within the images, and there is a lack of attention to multiscale features, resulting in insufficient utilization of details across different scales.

B. Augmented View-Based Contrastive Methods

Self-supervised learning through contrastive methods is also commonly applied in HSI classification tasks. These methods typically maximize the consistency between augmented views of the same sample in latent space and the differences between different samples. They have garnered significant attention, leading to a series of valuable studies [32], [33], [34], [35], [36], [37], [38]. For instance, Hou et al. [32] applied random noise to HSI patches to create augmented views, constructing positive and negative sample pairs for contrastive learning. Zhao et al. [33] designed an enhancement module specific to HSI dimensions, generating multiview samples to increase sample quantity and utilizing a self-supervised learning framework with a twin network structure to extract effective spatial and spectral representations through contrastive loss. Xia et al. [34] proposed a dual-branch contrastive learning network that enhances spatial and spectral features separately for spatial and spectral feature contrastive learning. Zhao et al. [35] proposed a contrastive learning-based method called ProCoL for HSI classification. ProCoL enhances the model's ability to learn image features by introducing an auxiliary low-dimensional subspace and a staged training strategy, which improves sensitivity to relationships between samples.

Although these contrastive learning methods based on enhanced views reduce the reliance on labeled samples for HSI classification and consider intersample relationships to some extent, they still have certain limitations in feature learning. These methods often rely on a single data augmentation strategy, limiting their ability to capture image features from a multiscale perspective. In addition, during the pretraining phase, they may focus on redundant features, without sufficiently capturing the deeper connections between spatial and spectral features. This limits the model's capacity to extract more discriminative features from the data, ultimately impacting classification accuracy and robustness.

C. Contrastive-Generation Cascaded Methods

Given the respective advantages of masked generative self-supervised methods and contrastive self-supervised learning methods, some researchers have attempted to combine both approaches for more effective HSI data modeling [39], [40], [41], [42]. For example, Cao et al. [39] combined contrastive learning and masked autoencoders, proposing a dual-branch masked autoencoder network based on contrastive learning. The first branch uses an encoder to capture latent

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representations of masked HSIs, while the decoder reconstructs pixel-level HSIs. The second branch, comprising a momentum encoder and projection head, embeds the image into feature space, enhancing the model's classification performance through contrastive learning. Huang et al. [40] designed a masked Transformer network using contrastive learning, where masked and unmasked views of the same HSI are used as positive and negative sample pairs for contrastive learning. In addition, Liang et al. [41] proposed a method called NeiCoT, which enhances the feature learning ability of Transformer models in HSI classification by combining masked reconstruction and neighborhood contrastive learning strategies. The neighborhood contrastive learning task maximizes the mutual information between local image patches and global average representations, improving the semantic richness and contextual sensitivity of the learned features.

Although these methods that combine masked generation and contrastive learning have made progress in HSI classification, they often do not fully consider the interrelationship between local features and the surrounding context in HSI data, particularly the connection between central pixels and neighboring feature information. Moreover, these methods tend to focus excessively on single-scale feature learning, neglecting the complementarity of features across different scales. In addition, there is a risk of paying excessive attention to less discriminative features during training, which can limit the model's generalization ability.

To address these issues, we designed a multiscale spatial–spectral masked Transformer network with LCL for HSI classification. Our method incorporates a multiscale spatial–spectral feature enhancement module to generate feature maps that blend information from different scales for masked reconstruction, enabling the model to leverage multiscale information during subsequent feature learning. We also developed a novel center-patch masking strategy to help the model capture deeper connections between central local features and surrounding spatial–spectral information. Furthermore, the designed LCL ensures the model focuses more on central information during reconstruction, effectively suppressing redundant features. In the fine-tuning stage, we introduced a multiscale adaptive feature fusion module to adaptively integrate multiscale information for final classification, further utilizing the complementarity of features across different scales.

III. Methodology

In this section, we will introduce the proposed MSMT-LCL method from the perspectives of self-supervised pretraining and supervised fine-tuning. The specific content includes the model architecture, the proposed MAFM, the center-patch masking strategy, the LCL strategy, and the AMFM.

A. Pretraining Stage

The proposed MSMT-LCL method's pretraining architecture is shown in Fig. 1. This method consists of three main components: the MAFM, the multiscale masked spatial–spectral Transformer (MSST), and the LCL component. The MAFM generates two types of mixed-scale feature maps through multiscale random cropping and feature mapping. The MSST employs a center-patch masking strategy for multiscale masked reconstruction. The LCL enables mutual learning between the two reconstruction branches during training, with a focus on modeling central local information.

1) MAFM Module: Fig. 2 illustrates the proposed MAFM module. This module generates patches of varying scales through multiscale cropping and random selection strategies. These patches are then processed via 2-D convolutions and residual connections to meticulously map and integrate spatial features and spectral information at different scales, ultimately producing a feature map that combines these multiscale inputs. By enriching the diversity of training samples, MAFM not only broadens the dataset but also enhances the network's ability to extract finer spatial and spectral details from HSI, thereby improving the model's feature learning capability.

Assume that $X \in \mathbb{R}^{H\times W\times S}$ represents a patch of the entire HSI, where H and W denote the height and width dimensions, respectively, and S represents the spectral dimension. The input patch X is cropped into five different scales using specified window sizes. Specifically, taking the spatial size $H \times W$ as a reference and using a as the interval, the patch is cropped down five times at equal intervals to generate five patches of different scales. One patch is then randomly selected from the two smallest patches and resized to the input size $H \times W$. Similarly, the same operation is applied to the two largest patches, and the middle-sized patch is resized to serve as the intermediate scale. This results in three patches P_L, P_M, and P_H, each containing different scale information.

Subsequently, these patches undergo a series of convolution operations, resulting in the mapped multiscale spatial–spectral features. To enhance feature fusion and improve the model's generalization ability, we use P_M as a residual connection feature, integrating it into the feature maps of different scales to obtain the output feature maps $F_L \in \mathbb{R}^{H\times W\times D}$ and $F_H \in \mathbb{R}^{H\times W\times D}$, where D is the spectral dimension after mapping. The MAFM module can be mathematically represented as follows:

$$\begin{aligned}(P_L, P_M, P_H) &= \mathrm{RSR}(\mathrm{MC}(X, a))\\ F_L &= \mathrm{Conv2D}(\mathrm{Conv2D}(P_L)) \oplus \mathrm{Conv2D}(P_M)\\ F_H &= \mathrm{Conv2D}(\mathrm{Conv2D}(P_H)) \oplus \mathrm{Conv2D}(P_M).\end{aligned} \tag{1}$$

Here, a denotes the specified interval size, MC represents multiscale cropping of the input X at intervals of a, and RSR denotes random selection and resizing of the cropped patches, resulting in $P_L, P_M, P_H \in \mathbb{R}^{H\times W\times S}$, representing the patches of low, medium, and high scales, respectively. The symbol $\oplus$ denotes element-wise addition for residual connections. F_L and $F_H \in \mathbb{R}^{H\times W\times D}$ represent the feature maps of low and high scales, respectively.

2) MSMT Reconstruction: The multiscale spatial–spectral Transformer reconstruction module includes two reconstruction branches, as shown in Fig. 1: the low-scale reconstruction branch and the high-scale reconstruction branch. Each branch

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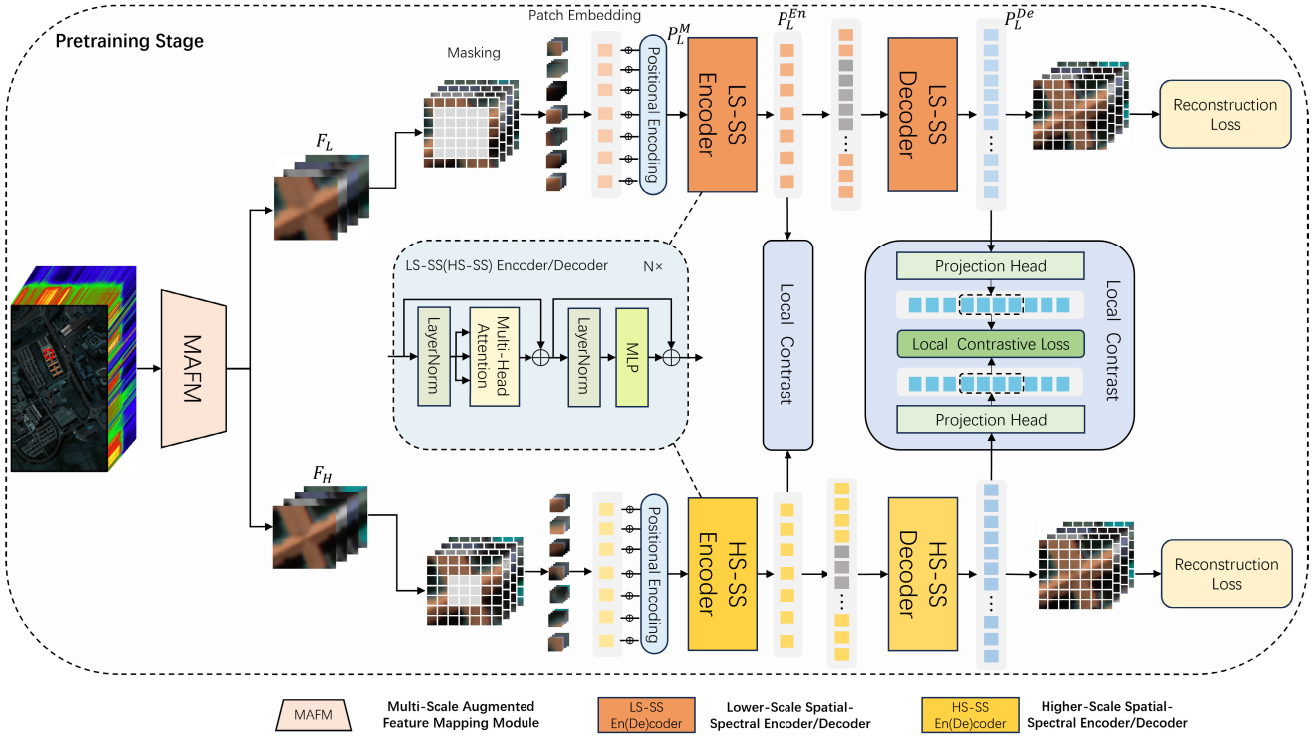


Fig. 1. Overall pretraining architecture of MSMT-LCL.

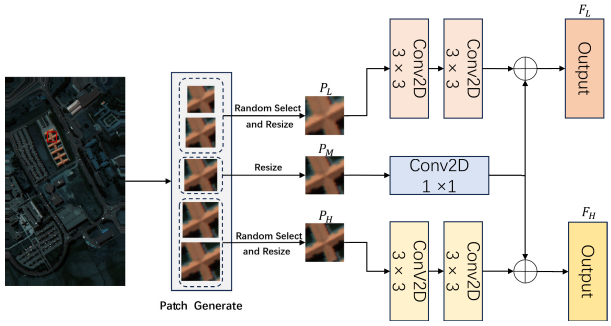


Fig. 2. Overall architecture of the proposed MAFM module.

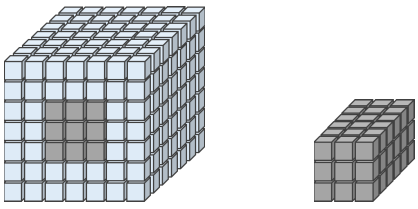


Fig. 3. Center-patch-based spatial–spectral masking strategy. (a) Spatial–spectral masking patch. (b) Spatial–spectral masking unit.

employs a center-patch masking strategy with different masking ratios to reconstruct the input HSI patch X. This approach allows the model to learn richer multiscale spatial–spectral information and the associations between central and surrounding information, enhancing the model's understanding of the data. In addition, we introduce LCL into the reconstruction branches to enhance mutual learning between the two branches, with a particular focus on modeling central local information.

Our masking strategy is illustrated in Fig. 3. It is well known that the central pixel and its local information are crucial for HSI classification, and the surrounding pixel information, which is often correlated with the central pixel, plays a key role in aiding the model's feature modeling. However, some existing random masking strategies can lead to redundant information modeling. To address this, we designed a center-patch-based spatial–spectral masking strategy. This strategy processes the central pixel and its surrounding area through masking, forcing the model to learn the potential associations between the central and surrounding features during reconstruction. Notably, we use different mask sizes in the two branches, enabling the model to learn at different scales and improve its understanding and modeling of multiscale features.

Taking the low-scale reconstruction branch as an example, the input feature map $F_L \in \mathbb{R}^{H\times W\times D}$. First, the input feature map is divided into a series of nonoverlapping patches through patch embedding and positional encoding, resulting in $F'_L \in \mathbb{R}^{N\times P^2D}$, where P is the patch size and $N = \mathrm{HW}/P^2$ is the number of patches. Then, a center-patch masking operation is applied to the patches based on a given mask window, resulting in $P_L^M \in \mathbb{R}^{T\times P^2D}$, where T denotes the masked dimension. This process can be mathematically represented as

$$P_L^M = \mathrm{CPM}\left(\mathrm{PE}(F_L) \oplus E_{\mathrm{pos}}, m\right) \tag{2}$$

$$E^i_{\mathrm{pos}}(j) = \begin{cases} \sin\left(\dfrac{i}{1000^{\frac{2j}{P^2D}}}\right), & j = 2n-1 \\ \cos\left(\dfrac{i}{1000^{\frac{2j}{P^2D}}}\right), & j = 2n \end{cases} \quad n = 1, 2, \cdots, \frac{P^2D}{2}. \tag{3}$$

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Here, PE(·) represents the patch embedding operation, E_{pos} denotes the positional encoding, CPM(·) represents the center-patch masking operation, and m denotes the specified mask window size.

Next, the masked data P_L^M is input into the Transformer-based low-scale spatial–spectral encoder for feature encoding to obtain features $P_L^{\text{En}} \in \mathbb{R}^{T\times P^2 D}$ that include long-range dependencies. The encoded features P_L^{En} are then passed through a linear mapping layer for feature transformation. Learnable vectors are filled into the masked positions, and positional encoding for reconstruction is added before inputting into the low-scale spatial–spectral decoder for reconstruction. The reconstruction output is then compared with the original input data to calculate the reconstruction loss. Notably, the encoder and the decoder at each layer have the same structure. The formula for the ith layer of the encoder and the decoder is as follows:

$$\begin{aligned} F_i' &= \text{MSA}(\text{LN}(F_{i-1})) + F_{i-1} \\ F_i &= \text{MLP}\big(\text{LN}\big(F_i'\big)\big) + F_i' \end{aligned} \tag{4}$$

where F_{i-1} represents the output of the $i-1$th layer of the encoder and the decoder, LN(·) represents layer normalization, MSA(·) represents multihead self-attention, and MLP(·) represents the multilayer perceptron. F_i represents the output of the ith layer.

In addition, we use mean squared error (mse) loss to compute the reconstruction losses for the two branches separately. The mathematical form is as follows:

$$\begin{aligned} L_{\text{ls}} &= \frac{1}{m_l}\sum_{i=1}^{m_l}(P_O - R_l)^2 \\ L_{\text{hs}} &= \frac{1}{m_h}\sum_{i=1}^{m_h}(P_O - R_h)^2 \\ L_{\text{rec}} &= L_{\text{ls}} + L_{\text{hs}} \end{aligned} \tag{5}$$

where L_{ls} and L_{hs} represent the reconstruction losses for the low-scale and high-scale reconstruction branches, respectively. We only compute the loss for the masked parts. m_l and m_h denote the number of masked patches in the two branches, R_l and R_h represent the reconstructed data for the two branches, and P_O represents the original input data after patch division. L_{rec} represents the total reconstruction loss.

3) Local Contrastive Learning: In the pretraining stage, we introduced an LCL mechanism that focuses on computing features near the central region. This approach allows the network to concentrate on capturing the most representative regional information, thereby reducing the influence of redundant features. This strategy significantly enhances the network's ability to model local central information during training and facilitates the mutual learning of the two branches in feature modeling. We used the encoder outputs from both branches as contrastive pairs, enabling the network to learn multiscale representations of the central pixels during feature modeling. To ensure that fine details of multiscale information are captured during both feature extraction and reconstruction, we also applied LCL to the decoder outputs of both branches. This further strengthens the network's ability to capture multiscale information and comprehend central local features during reconstruction. The LCL loss is computed using the NT-Xent loss [43].

For example, given the decoder outputs of the low-scale and high-scale reconstruction branches, F_L^{De} and F_H^{De}, respectively, the process can be mathematically expressed as

$$\begin{aligned} F_L^{\text{De}'} &= \text{PH}(F_L^{\text{De}}, c) \\ F_H^{\text{De}'} &= \text{PH}(F_H^{\text{De}}, c) \\ L_{\text{CDe}} &= \text{NT-Xent}(F_L^{\text{De}'}, F_H^{\text{De}'}, \tau) \end{aligned} \tag{6}$$

where PH(·) represents the projection head, consisting of a linear layer and an average pooling layer, and c represents the interval selection parameter for tokens used in LCL. $F_L^{\text{De}'}$ and $F_H^{\text{De}'}$ are the local features obtained after processing through the projection head. L_{CDe} represents the local contrastive loss for the decoder outputs and τ is the temperature parameter. The NT-Xent loss is mathematically defined as

$$\ell_{i,j} = -\log \frac{e^{s(x_i,x_j)/\tau}}{\sum_{k=1}^{2N} \mathbb{1}_{[k\neq i]} e^{s(x_i,x_j)/\tau}} \tag{7}$$

where N is the number of samples in a mini-batch, (x_i, x_j) represents the sample pair for the kth sample, and $s(p,q) = p^T q/\|p\|\|q\|$ denotes the dot product between the ℓ_2 normalized p and q. $\mathbb{1}_{[k\neq i]} \in \{0,1\}$ is an indicator function that returns 1 if $k \neq i$.

The same process is used to compute the local contrastive loss L_{CEn} for the encoder outputs from both branches. The overall loss of the model can be expressed as

$$\begin{aligned} L_{\text{con}} &= L_{\text{CEn}} + L_{\text{CDe}} \\ L &= L_{\text{rec}} + \lambda L_{\text{con}} \end{aligned} \tag{8}$$

where L_{con} represents the total contrastive loss, λ is the parameter that controls the weight of the LCL loss, and L is the total loss of the model.

B. Fine-Tuning Stage

After the unsupervised pretraining phase, the model possesses robust multiscale spatial–spectral feature modeling capabilities. Building on this foundation, we perform supervised fine-tuning, as illustrated in Fig. 4, to enable the model to accurately classify HSIs. During the fine-tuning stage, the masking operation and the decoder parts are removed. In addition, the multiscale augmentation operation of the MAFM module is eliminated, retaining only the feature mapping part.

To fully integrate the output features of the two branch encoders during fine-tuning, we designed the AMFM module, as depicted in Fig. 4. Specifically, let the outputs of the LS–SS encoder and HS–SS encoder be F_{LS} and F_{HS}, respectively. The process begins by independently applying convolution operations to F_{LS} and F_{HS}, resulting in F'_{LS} and F'_{HS}. Then, adaptive attention is computed for both sets of features through an adaptive average pooling layer and an additional convolutional layer, generating F''_{LS} and F''_{HS}. This step not only emphasizes the most relevant patterns at each scale but also highlights complementary information between features at different scales, further enhancing their discriminative ability. Finally, these two sets of features are fused to produce a

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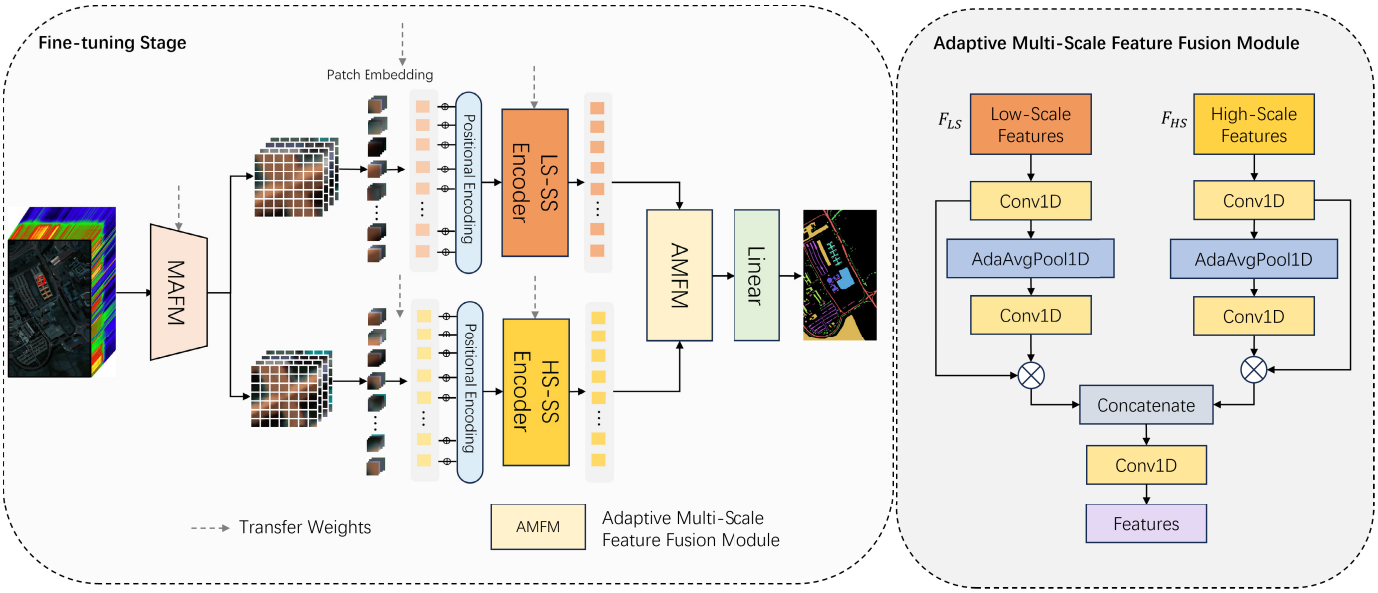


Fig. 4. Architecture of the fine-tuning and the AMFM module.

TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE PAVIA UNIVERSITY DATASET

Class	Name	Color	Training	Testing
1	Asphalt		20	6611
2	Meadows		20	18629
3	Gravel		20	2079
4	Trees		20	3044
5	Metal Sheets		20	1325
6	Bare Soil		20	5009
7	Bitumen		20	1310
8	Bricks		20	3662
9	Shadows		20	927
	Total		180	42596

unified output mapping. The mathematical representation of this module is as follows:

$$\begin{aligned} F'_{\mathrm{LS}} &= \mathrm{Conv1D}(F_{\mathrm{LS}}) \\ F'_{\mathrm{HS}} &= \mathrm{Conv1D}(F_{\mathrm{HS}}) \\ F''_{\mathrm{LS}} &= \mathrm{AvgPool}(\mathrm{Conv1D}(F'_{\mathrm{LS}})) \otimes F'_{\mathrm{LS}} \\ F''_{\mathrm{HS}} &= \mathrm{AvgPool}(\mathrm{Conv1D}(F'_{\mathrm{HS}})) \otimes F'_{\mathrm{HS}} \\ F &= \mathrm{Conv1D}(\mathrm{Concat}(F''_{\mathrm{LS}}, F''_{\mathrm{HS}})) \end{aligned} \tag{9}$$

where AvgPool denotes the adaptive average pooling layer, $\otimes$ represents element-wise multiplication, and Concat indicates the concatenation operation along a specified dimension. F represents the final output of the module. Lastly, the output F is passed through a linear layer, and the classification loss is computed using cross-entropy loss.

IV. EXPERIMENTS

In this section, we conducted a detailed experimental analysis on three public datasets: Pavia University, Indian Pines, and Houston2013. The analysis is organized into the following parts: data description, experimental setup, comparison of methods, ablation studies, and parameter experiments.

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE INDIAN PINES DATASET

Class	Name	Color	Training	Testing
1	Corn Notil		20	26
2	Corn Mintill		20	1408
3	Corn		20	810
4	Grass Pasture		20	217
5	Grass Trees		20	463
6	Hay Windrowed		20	710
7	Soybean Notill		20	8
8	Soybean Mintill		20	458
9	Soybean Clean		15	5
10	wheat		20	952
11	Woods		20	2435
12	Building Grass Trees Drives		20	573
13	Stone Steel Towers		20	185
14	Alfalfa		20	1245
15	Grass Pasture Mowed		20	366
16	Oats		20	73
	Total		315	9934

A. Datasets Description

1) Pavia University Dataset: This dataset was collected using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor by the University of Pavia, Italy. The spatial size of the dataset is 610 × 340 pixels, covering a spectral range of 430–860 nm, with a total of 103 spectral bands after removing 12 bands affected by water absorption. The dataset includes nine land-cover classes. Detailed class information and the number of training and testing samples used are provided in Table I.

2) Indian Pines Dataset: This dataset was collected using the AVIRIS sensor in the northwest region of Indiana, USA. The image has a spatial size of 145 × 145 pixels and covers a spectral range of 400–2500 nm, with a total of 200 spectral bands after removing 20 bands affected by water absorption.

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TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON2013 DATASET

Class	Name	Color	Training	Testing
1	Healthy Grass		20	1231
2	Stressed Grass		20	1234
3	Synthetic Grass		20	677
4	Tree		20	1224
5	Soil		20	1222
6	Water		20	305
7	Residential		20	1248
8	Commercial		20	1224
9	Road		20	1232
10	Highway		20	1207
11	Railway		20	1215
12	Parking Lot1		20	1213
13	Parking Lot2		20	449
14	Tennis Court		20	4008
15	Running Track		20	640
	Total		300	14729

The dataset includes 16 land-cover classes. Detailed class information and the number of training and testing samples used are provided in Table II.

3) Houston2013 Dataset: This dataset was collected using the CASI-1500 sensor in and around the University of Houston, USA. The spatial size of the dataset is 949 × 1905 pixels, and it includes 144 spectral bands ranging from 380 to 1050 nm. The version used in this study is the cloud-free version provided by the GRSS Data Fusion Contest [44]. The dataset includes 15 land-cover classes. Detailed class information and the number of training and testing samples used are provided in Table III.

B. Experimental Setup

Our proposed MSMT-LCL method was implemented using the PyTorch [45] deep learning framework and executed on an NVIDIA RTX 3080Ti GPU. All comparative methods were reimplemented using the publicly available code provided by the original authors and executed under the same hardware configuration for consistency. The Adam optimizer was employed for gradient descent, with an initial learning rate set to $1e^{-3}$. Both the pretraining and fine-tuning stages used a batch size of 512. The pretraining stage was run for 300 epochs, while the fine-tuning stage was run for 180 epochs.

To evaluate the classification performance of all comparative methods, we employed three metrics: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. To minimize the impact of random sampling on classification results, all experiments were repeated ten times, with the results reported as the mean ± standard deviation. Detailed counts of the training and testing samples used are listed in Tables I–III.

C. Comparison Methods

To validate the effectiveness of our proposed MSMT-LCL method, we compared it with several state-of-the-art methods on the three public datasets. These methods include RNN-based method [9], 2D-CNN-based method [46], the hybrid 2D-CNN and 3D-CNN method HybridSN [47], Transformer-based methods such as ViT [16], the combined CNN and Transformer method morphFormer [48], the contrastive learning-based masked Transformer method C-SS-MTr [40], the combined CNN and masked Transformer method CMT [49], and the spatial and spectral masked Transformer method SS-MAE [50]. It is important to note that all comparative methods were reimplemented using the publicly available code provided by the original authors, and the parameters used for each method followed the settings specified in the original papers. To ensure a fair comparison, the SS-MAE method was reproduced using only HSI images as input.

1) Quantitative Analysis: The quantitative analysis results on three datasets are presented in Tables IV–VI. These tables demonstrate that our proposed method outperforms all compared methods in terms of OA, AA, and the Kappa coefficient. Specifically, the OA results on the Pavia University, Indian Pines, and Houston 2013 datasets reached 94.49%, 88.78%, and 93.73%, respectively. Notably, the bold numbers in the tables indicate the best results for each category.

The experimental results on the Pavia University dataset (Table IV) show that the RNN-based method [9] achieved the worst performance, with an OA of only 73.50%. This poor performance is likely due to inadequate modeling of spatial information. In contrast, the 2D-CNN [46] and HybridSN [47] methods, which are convolution-based, improved OA by 12.64% and 7.69%, respectively, likely because CNNs have strong local spatial feature extraction capabilities. Although the ViT method [16] showed some advantages in modeling long-range dependencies, it performed poorly on high-dimensional HSI data, with an OA of only 74.90%, possibly due to insufficient utilization of spatial information. Conversely, the morphFormer method [48], which combines the strengths of CNNs and Transformers, improved the OA by 10.91% compared to the ViT method.

Given the scarcity of labeled training samples, the C-SS-MTr [40], CMT [49], and SS-MAE [50] methods employed a pretraining and fine-tuning strategy, achieving higher accuracy with fewer samples. Their OA results were 90.07%, 91.07%, and 91.64%, respectively, outperforming other methods that did not use pretraining. Our proposed MSMT-LCL method achieved the best results, with OA improvements of 4.42%, 3.42%, and 2.85% compared to the C-SS-MTr, CMT, and SS-MAE methods, respectively. The MSMT-LCL method's outstanding performance can be attributed to its integration of multiscale masked reconstruction learning with LCL, effectively capturing the complex structures and patterns within HSI data.

Similar conclusions can be drawn from Tables V and VI. For instance, in the class-imbalanced Indian Pines dataset (Table V), our MSMT-LCL method again achieved the best results, with OA improvements of 1.07% and 12.39% over the best pretrained method (CMT) and the CNN–Transformer hybrid method (morphFormer), respectively. In Table VI, our method outperformed the second-best method (SS-MAE) in

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TABLE IV
COMPARATIVE EXPERIMENTAL RESULTS ON THE PAVIA UNIVERSITY DATASET

Class	Methods								
#	RNN	2D-CNN	HybridSN	ViT	morphFormer	C-SS-MTr	CMT	SS-MAE	MSMT-LCL
1	57.51 ± 5.10	81.62 ± 4.05	71.00 ± 7.94	68.01 ± 3.80	85.56 ± 4.31	**93.49 ± 3.76**	91.95 ± 4.29	**93.49 ± 4.40**	92.29 ± 5.09
2	75.37 ± 5.34	83.24 ± 5.69	86.21 ± 6.52	70.52 ± 6.93	81.35 ± 11.04	85.37 ± 5.97	89.47 ± 4.87	87.06 ± 5.74	**92.27 ± 3.15**
3	66.39 ± 6.47	85.04 ± 6.39	65.86 ± 6.23	67.52 ± 7.87	72.63 ± 14.30	95.06 ± 2.92	88.74 ± 3.72	95.29 ± 3.44	**96.41 ± 2.43**
4	90.93 ± 3.34	96.19 ± 0.93	97.35 ± 1.16	94.32 ± 2.08	94.68 ± 1.85	88.46 ± 5.17	92.94 ± 2.35	96.99 ± 1.81	**98.10 ± 0.82**
5	99.74 ± 0.14	99.26 ± 0.57	99.77 ± 0.34	99.37 ± 0.76	98.69 ± 1.09	93.86 ± 3.47	99.43 ± 0.91	**99.95 ± 0.06**	99.87 ± 0.17
6	73.03 ± 6.56	87.95 ± 11.63	76.04 ± 4.86	77.51 ± 10.92	94.26 ± 3.20	96.50 ± 3.03	92.61 ± 8.95	94.74 ± 3.65	**98.63 ± 1.32**
7	79.93 ± 6.24	92.79 ± 2.33	78.11 ± 6.73	83.22 ± 6.72	89.66 ± 3.93	98.87 ± 1.71	98.71 ± 2.34	98.92 ± 1.46	**99.19 ± 0.93**
8	64.58 ± 4.85	88.64 ± 4.72	66.18 ± 8.16	76.20 ± 4.56	88.57 ± 4.02	**95.61 ± 3.99**	87.72 ± 8.43	93.18 ± 4.26	95.22 ± 1.96
9	**99.91 ± 0.09**	98.54 ± 0.84	99.41 ± 0.80	98.65 ± 0.77	97.04 ± 1.21	80.04 ± 7.19	98.31 ± 0.72	99.78 ± 0.40	99.35 ± 1.04
OA(%)	73.50 ± 2.32	86.14 ± 2.90	81.19 ± 3.80	74.90 ± 3.31	85.81 ± 5.41	90.07 ± 2.79	91.07 ± 2.31	91.64 ± 2.18	**94.49 ± 1.70**
AA(%)	78.60 ± 1.46	90.36 ± 1.94	82.21 ± 2.27	81.70 ± 2.35	89.16 ± 2.97	91.91 ± 1.77	93.32 ± 1.71	95.49 ± 0.78	**96.81 ± 0.96**
Kappa(%)	66.33 ± 2.59	82.16 ± 3.61	75.67 ± 4.56	68.39 ± 3.84	81.90 ± 6.55	87.21 ± 3.47	88.38 ± 2.94	89.19 ± 2.71	**92.82 ± 2.16**

TABLE V
COMPARATIVE EXPERIMENTAL RESULTS ON THE INDIAN PINES DATASET

Class	Methods								
#	RNN	2D-CNN	HybridSN	ViT	morphFormer	C-SS-MTr	CMT	SS-MAE	MSMT-LCL
1	86.54 ± 8.81	**100.00 ± 0.00**	89.74 ± 8.81	98.08 ± 3.10	98.46 ± 2.55	99.62 ± 1.15	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
2	36.55 ± 5.99	56.89 ± 5.78	32.20 ± 7.46	46.28 ± 5.03	62.22 ± 7.46	71.70 ± 4.56	**85.43 ± 5.44**	71.20 ± 7.21	77.03 ± 4.65
3	41.46 ± 5.30	71.21 ± 9.08	23.83 ± 11.54	62.33 ± 5.96	77.95 ± 8.35	82.14 ± 4.78	86.96 ± 4.23	84.00 ± 4.49	**89.48 ± 3.97**
4	50.78 ± 9.08	97.88 ± 2.66	52.07 ± 23.53	86.96 ± 7.11	95.16 ± 4.07	96.91 ± 3.40	96.41 ± 6.11	96.45 ± 2.65	**98.90 ± 1.65**
5	74.97 ± 4.26	86.37 ± 3.27	42.91 ± 26.70	75.62 ± 6.49	90.32 ± 3.63	85.33 ± 4.46	89.89 ± 3.34	90.43 ± 2.82	**92.98 ± 1.51**
6	88.54 ± 3.35	93.77 ± 4.29	51.91 ± 29.52	68.51 ± 7.60	94.76 ± 3.25	94.15 ± 7.42	96.80 ± 1.59	97.94 ± 1.45	**98.92 ± 1.10**
7	90.00 ± 5.00	**100.00 ± 0.00**	94.44 ± 6.59	98.75 ± 3.75	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
8	80.72 ± 5.24	99.63 ± 0.51	91.43 ± 7.23	96.42 ± 2.44	97.95 ± 1.29	99.61 ± 0.77	99.76 ± 0.23	99.63 ± 0.70	**100.00 ± 0.00**
9	92.00 ± 13.27	**100.00 ± 0.00**	97.78 ± 6.67	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
10	47.16 ± 7.31	68.35 ± 4.02	35.73 ± 22.39	52.65 ± 8.65	73.95 ± 6.30	80.82 ± 5.31	85.65 ± 5.02	84.64 ± 5.08	**89.79 ± 3.00**
11	37.10 ± 4.96	68.16 ± 4.19	37.43 ± 11.87	39.97 ± 7.15	62.04 ± 10.87	80.70 ± 5.54	78.27 ± 8.17	75.71 ± 5.69	**83.17 ± 5.16**
12	45.81 ± 5.16	69.39 ± 6.54	30.83 ± 14.35	54.26 ± 7.95	70.28 ± 10.20	78.80 ± 5.19	**83.14 ± 8.09**	80.45 ± 9.88	81.13 ± 6.28
13	96.59 ± 1.45	99.89 ± 0.22	82.94 ± 17.79	97.41 ± 4.63	99.30 ± 1.08	96.81 ± 4.27	99.41 ± 0.56	99.03 ± 1.90	**100.00 ± 0.00**
14	73.86 ± 6.42	90.63 ± 2.18	76.12 ± 9.84	75.17 ± 7.90	87.35 ± 5.48	95.36 ± 2.66	95.10 ± 3.27	**96.77 ± 1.89**	96.67 ± 1.56
15	57.21 ± 7.13	92.76 ± 4.12	49.48 ± 20.96	81.61 ± 5.25	92.10 ± 4.27	94.13 ± 3.50	**98.17 ± 1.68**	94.62 ± 3.35	98.14 ± 2.01
16	89.73 ± 5.42	99.73 ± 0.55	94.67 ± 10.56	95.75 ± 5.36	97.26 ± 3.73	96.03 ± 3.80	99.45 ± 0.91	**99.59 ± 0.63**	**99.59 ± 0.60**
OA(%)	53.64 ± 2.12	76.35 ± 1.21	45.91 ± 8.73	59.71 ± 2.23	76.39 ± 3.55	84.67 ± 2.05	87.71 ± 2.22	84.75 ± 2.10	**88.78 ± 1.72**
AA(%)	68.06 ± 1.44	87.17 ± 0.75	61.47 ± 9.84	76.86 ± 1.35	87.44 ± 1.55	90.76 ± 1.10	93.40 ± 0.78	91.90 ± 1.26	**94.11 ± 0.97**
Kappa(%)	48.13 ± 2.33	73.27 ± 1.32	39.50 ± 9.86	55.22 ± 2.31	73.45 ± 3.87	82.63 ± 2.29	86.08 ± 2.47	82.68 ± 2.36	**87.26 ± 1.93**

TABLE VI
COMPARATIVE EXPERIMENTAL RESULTS ON THE HOUSTON2013 DATASET

Class	Methods								
#	RNN	2D-CNN	HybridSN	ViT	morphFormer	C-SS-MTr	CMT	SS-MAE	MSMT-LCL
1	93.36 ± 3.26	91.98 ± 7.08	84.62 ± 6.51	78.43 ± 6.56	89.80 ± 6.44	87.24 ± 4.61	85.02 ± 7.76	**94.48 ± 4.68**	94.40 ± 5.04
2	96.75 ± 3.26	88.01 ± 6.59	85.74 ± 5.22	71.86 ± 10.87	92.76 ± 4.20	89.47 ± 4.11	92.98 ± 4.18	96.58 ± 1.53	**96.98 ± 4.53**
3	**100.00 ± 0.00**	98.08 ± 2.50	90.31 ± 8.12	93.60 ± 5.82	96.69 ± 5.33	97.99 ± 2.16	99.32 ± 0.34	99.91 ± 0.27	99.48 ± 0.40
4	95.51 ± 2.60	93.24 ± 2.69	90.33 ± 4.86	70.00 ± 9.26	93.47 ± 3.90	83.78 ± 11.31	96.04 ± 3.20	97.45 ± 2.09	**97.82 ± 1.71**
5	98.48 ± 1.05	98.89 ± 3.34	94.15 ± 2.57	93.18 ± 4.03	95.38 ± 5.37	97.30 ± 5.03	**99.83 ± 0.42**	99.21 ± 0.60	98.63 ± 3.96
6	95.67 ± 4.89	87.90 ± 4.23	89.08 ± 4.38	80.92 ± 7.23	93.15 ± 3.90	92.43 ± 4.13	**97.48 ± 3.60**	94.85 ± 4.46	94.10 ± 4.15
7	76.94 ± 8.80	89.52 ± 1.87	79.33 ± 4.99	72.56 ± 6.53	84.78 ± 2.80	89.45 ± 3.77	78.87 ± 5.34	87.32 ± 4.57	**90.54 ± 2.78**
8	50.47 ± 6.59	62.83 ± 8.00	63.90 ± 4.53	57.49 ± 4.35	75.83 ± 4.67	77.21 ± 8.14	66.26 ± 6.54	**79.32 ± 4.99**	80.04 ± 4.90
9	65.08 ± 15.46	85.34 ± 1.83	65.82 ± 6.67	55.88 ± 8.74	84.69 ± 6.86	86.44 ± 3.63	77.25 ± 8.97	83.74 ± 4.58	**92.95 ± 3.04**
10	70.20 ± 8.15	79.78 ± 7.99	63.21 ± 10.49	73.93 ± 8.40	86.63 ± 3.81	93.66 ± 6.04	90.56 ± 6.18	91.62 ± 4.98	**94.43 ± 3.88**
11	75.09 ± 7.01	75.70 ± 9.51	69.18 ± 7.44	71.79 ± 8.89	78.82 ± 5.53	**91.97 ± 4.36**	87.05 ± 7.38	84.64 ± 7.67	91.37 ± 5.56
12	59.88 ± 9.15	80.54 ± 9.01	66.20 ± 5.44	60.85 ± 7.75	82.67 ± 7.57	81.26 ± 8.26	74.75 ± 11.50	86.64 ± 7.46	**90.58 ± 5.88**
13	52.47 ± 9.83	94.10 ± 1.90	92.92 ± 2.95	70.18 ± 7.94	89.84 ± 4.80	93.21 ± 4.59	86.30 ± 2.66	94.50 ± 3.24	**96.39 ± 2.55**
14	98.92 ± 0.70	99.53 ± 1.16	97.08 ± 4.31	97.01 ± 2.32	98.46 ± 1.66	**100.00 ± 0.00**	**100.00 ± 0.00**	99.95 ± 0.15	**100.00 ± 0.00**
15	99.02 ± 0.30	99.64 ± 1.08	96.34 ± 2.86	85.27 ± 5.74	98.45 ± 2.76	98.14 ± 1.45	99.94 ± 0.19	99.97 ± 0.06	**100.00 ± 0.00**
OA(%)	80.27 ± 1.20	86.66 ± 1.57	79.15 ± 1.72	73.22 ± 2.58	88.05 ± 1.56	89.30 ± 1.64	86.89 ± 1.18	91.49 ± 1.36	**93.73 ± 1.52**
AA(%)	81.86 ± 1.28	88.34 ± 1.26	81.88 ± 1.34	75.53 ± 2.40	89.43 ± 1.19	90.64 ± 1.38	88.78 ± 1.03	92.68 ± 1.14	**94.51 ± 1.14**
Kappa(%)	78.67 ± 1.30	85.57 ± 1.70	77.49 ± 1.86	71.13 ± 2.77	87.09 ± 1.68	88.43 ± 1.77	85.84 ± 1.27	90.80 ± 1.47	**93.23 ± 1.64**

terms of OA, AA, and Kappa by 2.24%, 1.83%, and 2.43%, respectively. Our method's exceptional performance is likely due to the adoption of a center-patch-based masking strategy and LCL techniques. This approach allows the model to focus more intently on the intrinsic connections between central local information and surrounding contextual information during the modeling process, thereby more effectively capturing key features within the data.

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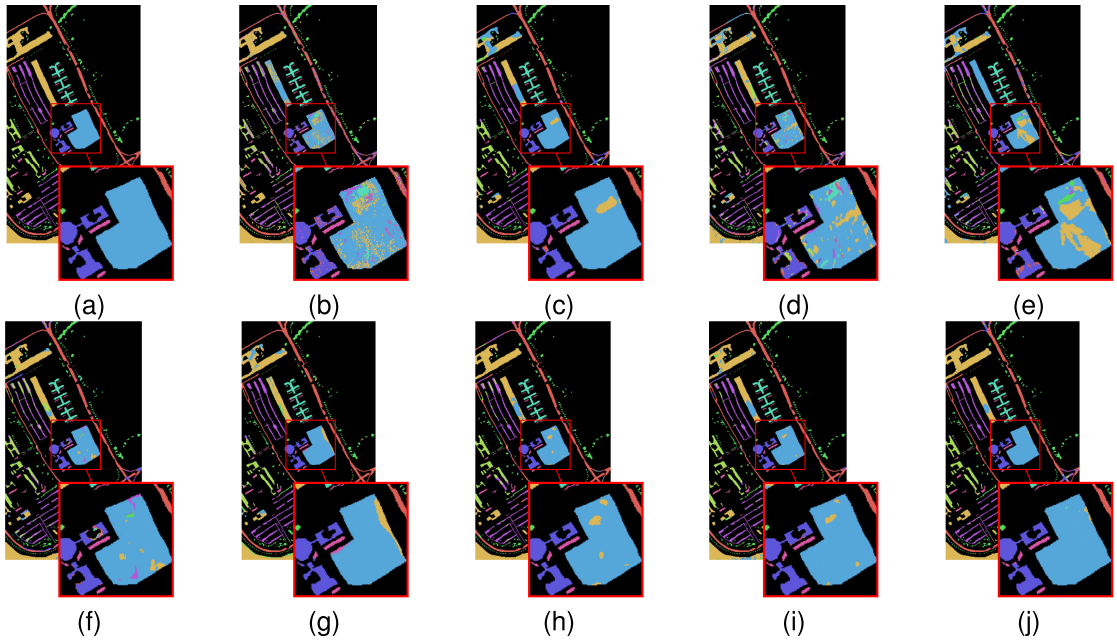


Fig. 5. Visualization of classification maps on the Pavia University dataset. (a) Ground truth, (b) RNN, (c) 2D-CNN, (d) HybridSN, (e) ViT, (f) morphFormer, (g) C-SS-MTr, (h) CMT, (i) SS-MAE, and (j) MSMT-LCL (Ours).

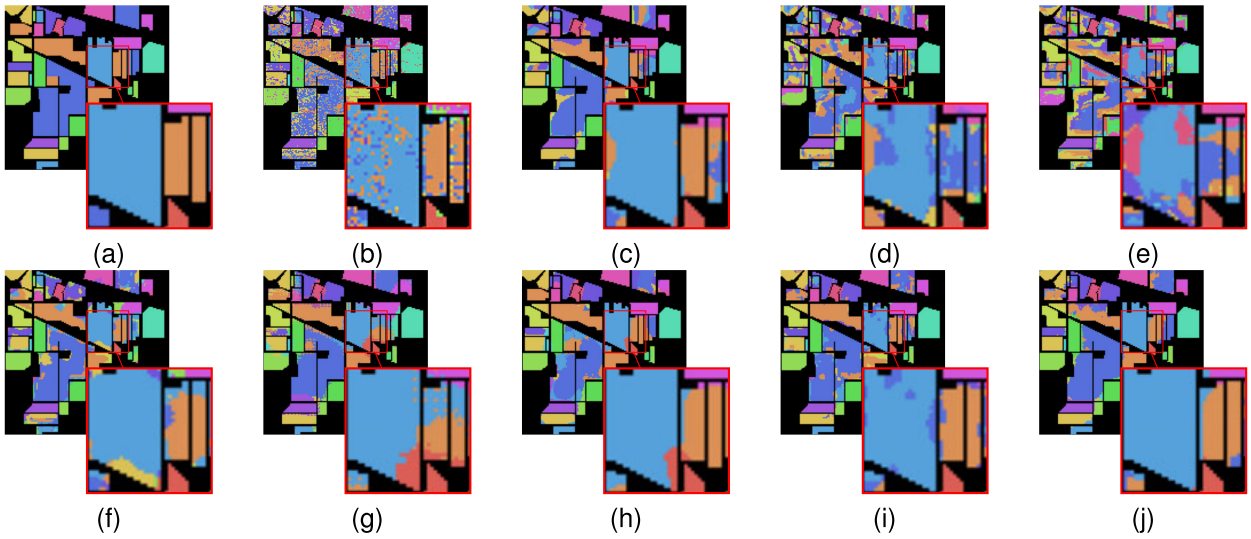


Fig. 6. Visualization of the classification maps based on the Indian Pines dataset. (a) Ground truth, (b) RNN, (c) 2D-CNN, (d) HybridSN, (e) ViT, (f) morphFormer, (g) C-SS-MTr, (h) CMT, (i) SS-MAE, and (j) MSMT-LCL (Ours).

2) Qualitative Analysis: To better analyze the experimental results qualitatively, we conducted visualization experiments on the three datasets, as shown in Figs. 5–7. To clearly observe the differences in visual feature maps produced by different methods, we enlarged the local images within the red rectangles.

As shown in Fig. 5, the visual results on the Pavia University dataset indicate that our proposed method produces classification maps with the least noise and is closest to the ground truth. In contrast, other methods, such as morphFormer, exhibit more misclassifications. This may be because, with limited training samples, supervised learning alone is insufficient to train a high-performance classifier. In addition, although the C-SS-MTr method uses a masked Transformer pretrained with contrastive learning, the classification maps still contain significant noise, possibly due to the introduction of redundant information during modeling, which interferes with the classification results. Our proposed method mitigates the modeling of redundant information through the designed center-patch masking strategy and LCL, making it easier for the model to capture important features, which may be one reason for achieving better classification results.

As shown in Figs. 6 and 7, our proposed method achieves the best visual results on both the Indian Pines and Houston 2013 datasets, producing the least noise and aligning most closely with the ground truth. For example, in Fig. 7, the methods by RNN and ViT generate excessive noise, likely due to insufficient spatial information modeling, which prevents them

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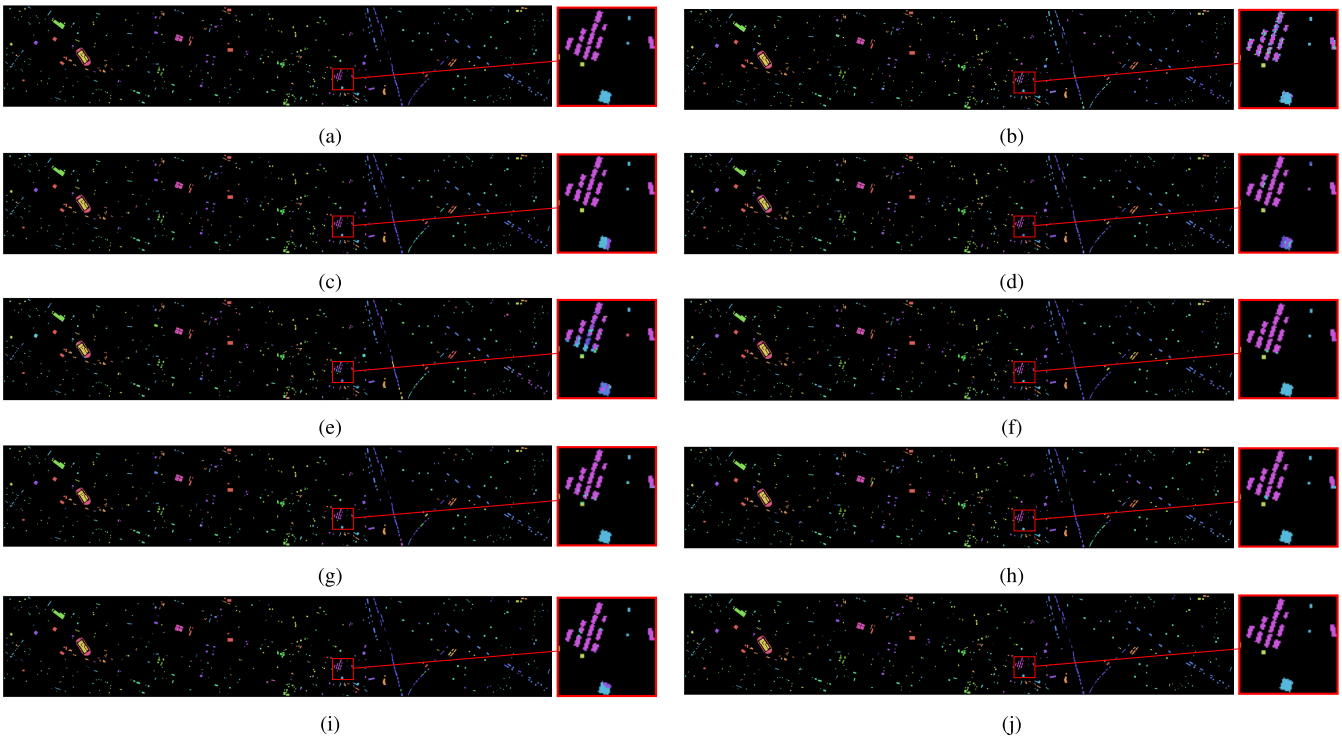


Fig. 7. Visualization of the classification maps based on the Houston2013 dataset. (a) Ground truth, (b) RNN, (c) 2D-CNN, (d) HybridSN, (e) ViT, (f) morphFormer, (g) C-SS-MTr, (h) CMT, (i) SS-MAE, and (j) MSMT-LCL (Ours).

from capturing the complex structural relationships between image pixels. In contrast, pretraining and fine-tuning methods such as C-SS-MTr, CMT, and SS-MAE show improved performance, likely due to their ability to learn rich feature representations during the pretraining phase, allowing for better adaptation to specific tasks during fine-tuning. However, even these methods show some degree of noise when compared to our method. The superior performance of our method is likely due to its use of a multiscale feature enhancement module, which captures information at multiple scales, and an AMFM during fine-tuning, enabling more effective integration of features at different scales and leading to more accurate classification results.

3) Complexity Analysis: As shown in Table VII, we assess the complexity of various methods on the Pavia University dataset using four metrics: floating-point operations (FLOPs), the number of parameters (Param), training time, and testing time. To ensure a fair comparison, methods based on pretraining and fine-tuning strategies are evaluated solely during the fine-tuning phase. All results are averaged over ten runs under consistent experimental conditions.

The table shows that our proposed method has a parameter size of 1.39 MB, placing it at a moderate level among the compared approaches, with reductions of 3.25 and 1.10 MB compared to SS-MAE and ViT, respectively. The training time for our method is 20.05 s, which is 4.47 and 16.51 s shorter than morphFormer and Hybridsn, respectively, indicating strong performance in this aspect. However, our method is less efficient in terms of FLOPs and testing time, which are 0.47 MB and 8.84 s, respectively. This phenomenon may be attributed to MSMT-LCL's in-depth modeling of multi-scale spatial–spectral features, which requires the network to perform detailed feature fusion and context analysis during learning to fully leverage information across different scales. This process not only increases the number of floating-point operations but also extends the testing time.

TABLE VII
COMPUTATIONAL COMPLEXITY ANALYSIS OF VARIOUS METHODS

Methods	FLOPs(GB)	Param(MB)	Training time(s)	Testing time(s)
RNN	0.002	0.09	10.59	1.14
2DCNN	0.02	0.43	5.76	2.32
HybridSN	0.26	2.53	36.56	7.83
ViT	0.19	2.49	20.18	6.25
morphFormer	0.02	0.15	24.52	7.74
C-SS-MTr	0.06	0.44	12.13	7.67
CMT	0.02	0.1	9.19	1.66
SS-MAE	0.21	4.64	16.4	2.43
MSMT-LCL	0.47	1.39	20.05	8.84

In summary, while MSMT-LCL incurs higher computational costs, it delivers significant improvements in classification accuracy compared to other methods, justifying the increased resource consumption. Future work will focus on optimizing the computational efficiency of MSMT-LCL to reduce training and testing times while maintaining its classification performance advantage.

D. Ablation Experiments

To validate the effectiveness of the proposed components, we conducted comprehensive experiments on three public datasets: Pavia University, Indian Pines, and Houston2013. Specifically, we performed ablation studies on the MAFM and AMFM modules, the LCL strategy, the center-patch masking

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TABLE VIII
ABLATION EXPERIMENT OF THE PROPOSED MAFM AND AMFM MODULES ON THREE DATASETS WITH OA RESULTS

Modules	MAFM	×	✓	✓
	AMFM	×	×	✓
Datasets	Pavia University	82.64±2.33	91.77±2.70	**94.49±1.70**
	Indian Pines	67.38±3.31	85.12±3.75	**88.78±1.72**
	Houston2013	77.68±2.70	91.51±2.93	**93.73±1.52**

TABLE IX
ABLATION EXPERIMENT OF THE PROPOSED LOCAL CONTRAST LEARNING STRATEGY ON THREE DATASETS WITH OA RESULTS

Datasets	Un-Contrast	Contrast	Local Contrast
Pavia University	92.98 ± 1.78	93.97 ± 1.37	**94.49 ± 1.70**
Indian Pines	86.67 ± 1.67	87.67 ± 2.24	**88.78 ± 1.72**
Houston2013	91.22 ± 1.92	93.12 ± 1.51	**93.73 ± 1.52**

TABLE X
ABLATION EXPERIMENT OF THE PROPOSED CENTER-PATCH MASKING STRATEGY ON THREE DATASETS WITH OA RESULTS

Datasets	Spectral Masking	Spatial Masking	Center-Patch Masking
Pavia University	80.32 ± 3.35	91.92 ± 2.24	**92.98 ± 1.78**
Indian Pines	76.52 ± 2.11	83.51 ± 2.15	**86.67 ± 1.67**
Houston2013	76.92 ± 2.78	90.53 ± 1.59	**91.22 ± 1.92**

TABLE XI
ABLATION EXPERIMENT OF THE PRETRAINING STAGE ON THREE DATASETS WITH OA RESULTS

Datasets	Un-Pretrained	Pretrained
Pavia University	91.03 ± 2.15	**94.49 ± 1.70**
Indian Pines	84.94 ± 2.12	**88.78 ± 1.72**
Houston2013	90.78 ± 2.03	**93.73 ± 1.52**

strategy, and the pretraining architecture. To mitigate the impact of random sampling on the experimental results, each experiment was repeated ten times, with results reported as mean ± standard deviation.

1) Ablation Experiment With MAFM and AMFM Modules: Table VIII presents the classification results obtained by including or excluding the MAFM and AMFM modules in our proposed method. Without these two modules, the model's OA was significantly lower, achieving only 82.64%, 67.38%, and 77.68% on the three datasets, respectively. When the MAFM module was included, the OA improved by 9.13%, 17.73%, and 13.93% on the respective datasets. This demonstrates that multiscale augmentation and shallow feature mapping are crucial for capturing multiscale details, which are vital for subsequent feature processing.

Furthermore, incorporating the AMFM module on top of the MAFM module led to additional OA improvements of 2.72%, 3.66%, and 2.22% on the three datasets, respectively. These results indicate that the AMFM plays a significant role in enhancing model performance. The AMFM module dynamically adjusts and fuses features at different scales, enabling the model to better capture the essential information needed for accurate classification.

2) Ablation Experiment With Local Contrast Learning Strategy: Table IX presents the ablation results for our proposed local contrast learning strategy. The table shows the OA results for three datasets under three conditions: without contrast learning, with conventional contrast learning [43], and with our proposed local contrast learning. When contrast learning is not used, the accuracies for the three datasets are 92.98%, 86.67%, and 91.22%, respectively. With conventional contrast learning, the accuracies increase by 0.99%, 1.00%, and 1.22%, respectively. When using our proposed local contrast learning, the accuracies further improve by 1.51%, 2.11%, and 2.51% compared to not using contrast learning. These results demonstrate that incorporating contrast learning during the pretraining stage enhances model performance. Specifically, conventional contrast learning improves the model's generalization ability by increasing the distinction between samples. Our proposed local contrast learning strategy further refines this process by focusing more on the contrast between central local features, enabling the model to better capture fine details and important characteristics.

3) Ablation Experiment for Center-Patch Masking Strategy: In reconstruction-based pretraining tasks, designing an appropriate masking strategy is crucial. To validate the effectiveness of our proposed center-patch masking strategy, we compared it with several common masking strategies, such as random spectral masking [27] and random spatial masking [40]. Table X presents the OA results from ablation experiments on three datasets using our center-patch masking strategy. To avoid the influence of local contrast learning on different masking strategies, this set of experiments did not use local contrast learning. Instead, we employed various masking reconstruction strategies for pretraining while keeping the rest of the network consistent with our proposed method.

The results in Table X show that our center-patch masking strategy achieved the best performance. For instance, on the class-imbalanced Indian Pines dataset, the OA improved by 10.15% and 3.16% compared to spectral masking and spatial masking strategies, respectively. This indicates that spectral masking leads to inadequate modeling of local spatial features, making the reconstruction task more challenging. On the other hand, spatial masking introduces a significant amount of redundant information, which does not generalize well. Our center-patch masking strategy focuses on the reconstruction of central local features, allowing the model to learn the relationship between central and surrounding features during the reconstruction process, thereby enhancing the model's generalization capability.

4) Ablation Experiment With Pretraining Stage: Table XI presents the results of the ablation experiment on the pretraining phase across three datasets, showing OA results. The table compares two scenarios: learning the classification task without pretraining and performing classification task fine-tuning after our designed pretraining phase. The results indicate that without pretraining, the model's accuracy drops by 3.46%, 3.84%, and 2.95% on the respective datasets. This demonstrates that the pretraining phase plays a critical role in enhancing model performance. Before supervised learning, utilizing self-supervised training helps the network learn richer and more representative feature embeddings, thereby

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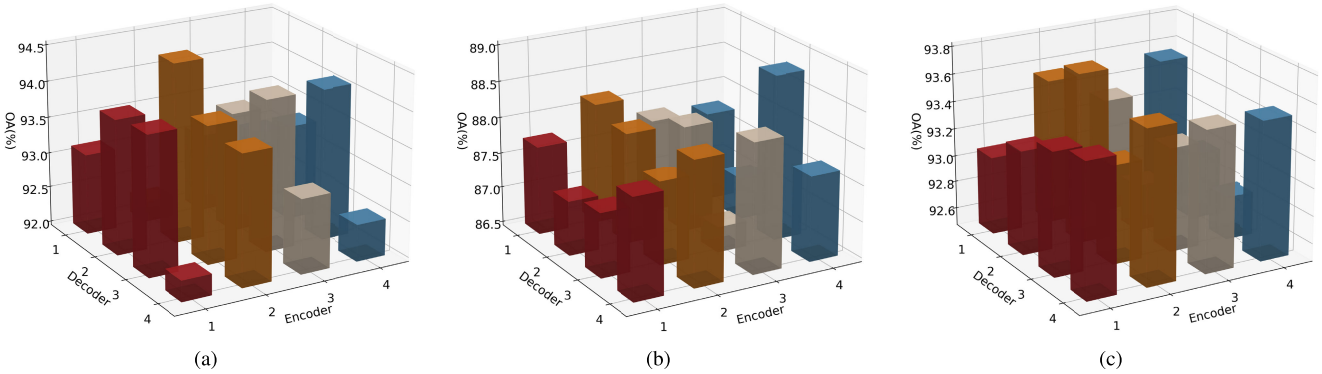


Fig. 8. Effect of the encoder and the decoder depth on OA results for different datasets. (a) Pavia University, (b) Indian Pines, and (c) Houston2013.

improving the model's understanding and adaptation to the data. Consequently, this increases the model's robustness and generalization capability in classification tasks.

E. Parameter Experiments

In this section, we analyze several hyperparameters that affect classification performance, including model depth, LCL loss parameters, input image patch size, and the size of masked patches during pretraining. We will discuss these experimental results in detail below.

1) Effect of Model Depth: The encoder and the decoder are crucial components of the model, playing a key role in feature extraction and significantly influencing classification results. We examined the impact of varying the number of layers in the encoder and the decoder on classification results. It is important to note that our high-scale and low-scale reconstruction branches use the same number of encoder and decoder layers. As shown in Fig. 8, we considered candidate layer numbers for the encoder and the decoder of $\{1, 2, 3, 4\}$.

The results indicate that for the Pavia University and Houston datasets, the model achieves optimal performance with lightweight encoders and decoders, specifically with $\{2, 2\}$ layers. For the Indian Pines dataset, the best performance is achieved with $\{4, 3\}$ layers in the encoder and the decoder, respectively. This suggests that more complex datasets require deeper models to effectively capture and model features.

2) Effect of Local Contrast Learning Loss Coefficient λ: This section analyzes the impact of different local contrast learning loss coefficients λ on classification results. As shown in Fig. 9, an inappropriate λ value, whether too large or too small, negatively affects the classification results. A small λ diminishes the effect of local contrast learning, making the network overly reliant on reconstruction loss, resulting in undesirable classification performance. Conversely, a large λ diminishes the importance of reconstruction loss, also leading to suboptimal classification results. Therefore, selecting an appropriate λ coefficient is crucial. The figure indicates that when λ is set to 0.05, the model achieves the best classification performance across all three datasets.

3) Effect of Input Patch Sizes: This section analyzes the impact of different input patch sizes on the model's classification performance. The candidate set includes $\{9 \times 9, 15 \times 15, 21 \times 21, 27 \times 27, 33 \times 33\}$. Given that the model performs

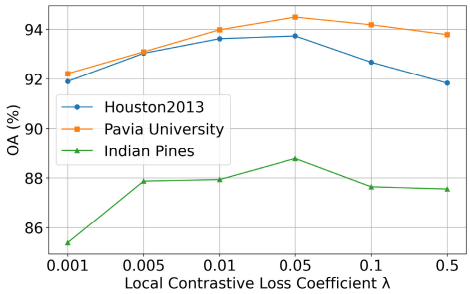


Fig. 9. Effect of different local contrast learning loss coefficients on OA results for different datasets.

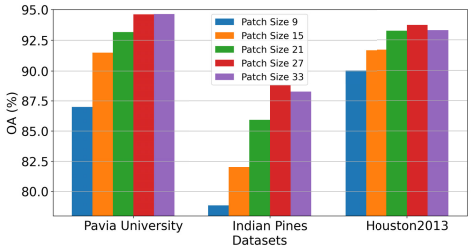


Fig. 10. Effect of different input patch sizes on OA results for different datasets.

multiscale augmentation for each patch size, we selected larger patch size intervals to comprehensively evaluate their effect on model performance.

As shown in Fig. 10, the model achieves the best results on the Indian Pines and Houston datasets with a patch size of 27×27. Smaller patch sizes result in reduced learning of relevant features, adversely affecting classification performance. Conversely, larger patch sizes introduce redundant information, interfering with the feature extraction process and leading to suboptimal results. On the Pavia University dataset, the model performs similarly with patch sizes of 27×27 and 33×33. This indicates that for this dataset, the model effectively extracts key features with both patch sizes. However, considering computational cost and efficiency, we ultimately chose 27×27 as the optimal size. This choice balances classification performance with computational resource utilization.

4) Effect of Center Patch Mask Size: Selecting the appropriate mask size during reconstruction is crucial, as sizes that are too large or too small can adversely affect the

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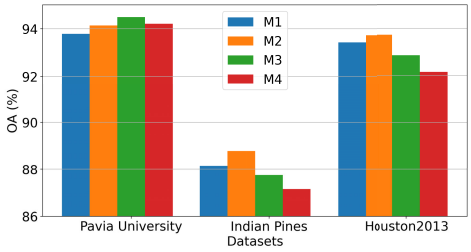


Fig. 11. Effect of different center patch mask sizes on OA results for different datasets.

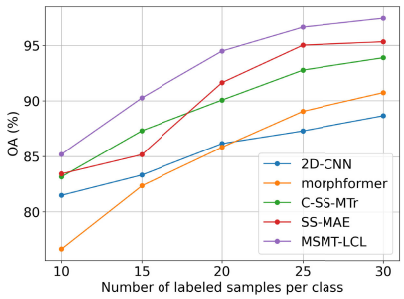


Fig. 12. Comparison of classification accuracy with varying numbers of training samples per class on the Pavia University dataset.

model's classification performance. This section investigates the impact of different mask sizes based on the center-patch mask strategy. To enhance the model's generalization ability, we employed different mask sizes in the high-scale reconstruction branch (B_H) and the low-scale reconstruction branch (B_L). The specific candidate sets are as follows: M1: (B_H : 3×3, B_L : 9×9), M2: (B_H : 9×9, B_L : 15×15), M3: (B_H : 15×15, B_L : 21×21), and M4: (B_H : 21×21, B_L : 24×24).

Our experimental results are shown in Fig. 11. From the figure, it is evident that an overly large mask size complicates the reconstruction task, negatively affecting classification performance. Conversely, an overly small mask size makes the reconstruction task too simple, leading to overfitting and suboptimal classification results. Specifically, the optimal mask patch sizes for the Pavia University, Indian Pines, and Houston2013 datasets are M3, M2, and M2, respectively.

These experimental results demonstrate that choosing an appropriate mask size not only effectively balances the complexity of the reconstruction task but also enhances the model's generalization ability and classification performance. Therefore, the rational design of mask size is a crucial step in improving model performance.

F. Assessing Results With Varied Training Samples

To further assess the scalability and generalization ability of the proposed method, we compared the OA results of different methods with varying numbers of training samples on the Pavia University dataset, as shown in Fig. 12. All results are averaged over ten runs conducted in a consistent experimental environment.

As illustrated in the figure, our proposed MSMT-LCL method consistently outperforms the compared methods across different training sample sizes, demonstrating its superior scalability and generalization capabilities.

V. Conclusion

In this work, we propose a novel method named MSMT-LCL for HSI classification. This method adopts a pretraining and fine-tuning strategy. Initially, the input data undergoes multiscale augmentation and feature mapping through the designed MAFM module, generating feature maps that blend different scales. These feature maps are then fed into low- and high-scale reconstruction branches. In addition, we introduce a center-patch-based masking strategy to enable the model to learn the relationship between central local information and surrounding information during the reconstruction process. In the pretraining phase, we incorporate an LCL strategy, utilizing features from encoders and decoders of the two reconstruction branches to enhance the model's focus on local central information across different scales. Following pretraining, the network undergoes supervised fine-tuning for the classification task. During fine-tuning, our designed AMFM module facilitates adaptive multiscale information fusion to improve the model's ability to learn diverse feature information. We conduct extensive experiments on three publicly available datasets: Pavia University, Indian Pines, and Houston2013, demonstrating that our proposed method achieves superior performance across all datasets.

In our future work, we aim to further explore and design new pretraining tasks and strategies to enhance the model's accuracy and generalization capabilities. Moreover, we will investigate methods to achieve better classification results using fewer samples and lighter-weight networks.

References

[1] G. Vivone, "Multispectral and hyperspectral image fusion in remote sensing: A survey," *Inf. Fusion*, vol. 89, pp. 405–417, Jan. 2023.

[2] S. Huang, H. Zhang, and A. Pižurica, "Subspace clustering for hyperspectral images via dictionary learning with adaptive regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524017.

[3] H. Gao, Y. Zhang, Z. Chen, F. Xu, D. Hong, and B. Zhang, "Hyperspectral target detection via spectral aggregation and separation network with target band random mask," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515516.

[4] S. Huang, H. Zhang, and A. Pižurica, "A structural subspace clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509515.

[5] C. Wang et al., "A review of deep learning used in the hyperspectral image analysis for agriculture," *Artif. Intell. Rev.*, vol. 54, no. 7, pp. 5205–5253, Oct. 2021.

[6] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[7] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, Aug. 2021.

[8] H. Gao, R. Sheng, Z. Chen, H. Liu, S. Xu, and B. Zhang, "Multiscale random-shape convolution and adaptive graph convolution fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516017.

[9] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[10] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.

[11] H. Gao, H. Feng, Y. Zhang, S. Xu, and B. Zhang, "AMSSE-Net: Adaptive multiscale spatial–spectral enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531317.

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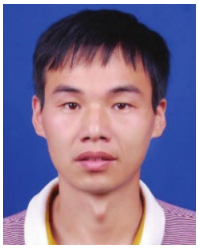
[12] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.
[13] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.
[14] C. Yu, R. Han, M. Song, C. Liu, and C.-I. Chang, "A simplified 2D-3D CNN architecture for hyperspectral image classification based on spatial–spectral fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2485–2501, 2020.
[15] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inform. Process. Syst. (NIPS)*, 2017, pp. 5998–6008.
[16] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–12.
[17] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Sep. 2019.
[18] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.
[19] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.
[20] X. Zhou, W. Zhou, X. Fu, Y. Hu, and J. Liu, "MDvT: Introducing mobile three-dimensional convolution to a vision transformer for hyperspectral image classification," *Int. J. Digit. Earth*, vol. 16, no. 1, pp. 1469–1490, 2023.
[21] J. Zhang, Z. Meng, F. Zhao, H. Liu, and Z. Chang, "Convolution transformer mixer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[22] Y. Zhou, X. Huang, X. Yang, J. Peng, and Y. Ban, "DCTN: Dual-branch convolutional transformer network with efficient interactive self-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508616.
[23] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.
[24] D. Ibañez, R. Fernandez-Beltran, F. Pla, and N. Yokoya, "Masked auto-encoding spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542614.
[25] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Self-supervised learning with prediction of image scale and spectral order for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5545715.
[26] L. Scheibenreif, M. Mommert, and D. Borth, "Masked vision transformers for hyperspectral image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 2166–2176.
[27] W. Liu et al., "Self-supervised feature learning based on spectral masking for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407715.
[28] Y. Chen and Q. Yan, "LFSMIM: A low-frequency spectral masked image modeling method for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[29] H. Liu, M. Zhang, Z. Di, M. Gong, T. Gao, and A. K. Qin, "A hybrid multitask learning network for hyperspectral image classification with few labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505816.
[30] L. Tu, J. Li, X. Huang, J. Gong, X. Xie, and L. Wang, "S2HM2: A spectral–spatial hierarchical masked modeling framework for self-supervised feature learning and classification of large-scale hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517019.
[31] F. Zhou, C. Xu, G. Yang, R. Hang, and Q. Liu, "Masked spectral–spatial feature prediction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4400913.
[32] S. Hou, H. Shi, X. Cao, X. Zhang, and L. Jiao, "Hyperspectral imagery classification based on contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521213.
[33] L. Zhao, W. Luo, Q. Liao, S. Chen, and J. Wu, "Hyperspectral image classification with contrastive self-supervised learning under limited labeled samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[34] S. Xia, X. Zhang, H. Meng, J. Fan, and L. Jiao, "Two-stream networks for contrastive learning in hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1903–1920, 2024.
[35] L. Zhao, Y. Feng, Y. Dai, J. Wu, and G. Zhang, "Progressive contrastive learning based on noisy negatives cleaning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[36] P. Guan and E. Y. Lam, "Cross-domain contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528913.
[37] B. Li, L. Fang, N. Chen, J. Kang, and J. Yue, "Enhancing hyperspectral image classification: Leveraging unsupervised information with guided group contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504317.
[38] L. Liu, H. Zhang, and Y. Wang, "Contrastive mutual learning with pseudo-label smoothing for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–14, 2024.
[39] X. Cao, H. Lin, S. Guo, T. Xiong, and L. Jiao, "Transformer-based masked autoencoder with contrastive loss for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524312.
[40] L. Huang, Y. Chen, and X. He, "Spectral–spatial masked transformer with supervised and contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023,Art. no. 5508718.
[41] M. Liang et al., "An efficient transformer with neighborhood contrastive tokenization for hyperspectral images classification," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 131, Jul. 2024, Art. no. 103979.
[42] M. Zhu, H. Wang, Y. Meng, Z. Shan, and Z. Ma, "Lightweight multiview mask contrastive network for small-sample hyperspectral image classification," in *Pattern Recognition and Computer Vision*. Singapore: Springer, 2024, pp. 478–490.
[43] T. Chen, S. Kornblith, M. Norouzi, and G. E. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. 37th Int. Conf. Mach. Learn.*, vol. 119, 2020, pp. 1597–1607.
[44] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.
[45] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. NIPS*, Dec. 2019, pp. 8024–8035.
[46] V. Sharma, A. Diba, T. Tuytelaars, and L. Van Gool, "Hyperspectral CNN for image classification & band selection, with application to face recognition," KU Leuven, ESAT, Leuven, Belgium, Tech. Rep. KUL/ESAT/PSI/1604, 2016.
[47] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2019.
[48] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.
[49] S. Jia, Y. Wang, S. Jiang, and R. He, "A center-masked transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024,Art. no. 5510416.
[50] J. Lin, F. Gao, X. Shi, J. Dong, and Q. Du, "SS-MAE: Spatial–spectral masked autoencoder for multisource remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531614.



Yunfei Zhou received the B.S. degree from Zaozhuang University, Zaozhuang, China, in 2021. He is currently pursuing the M.S. degree in computer technology with East China Jiaotong University, Nanchang, China.

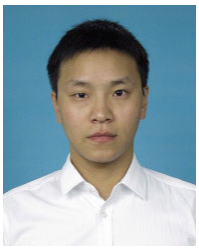
He is currently a Researcher with the Institute for Data Science and Deep Learning, East China Jiaotong University. His research interests include deep learning and hyperspectral image classification.

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Xiaohui Huang (Member, IEEE) received the B.Eng. and master's degrees from Jiangxi Normal University, Nanchang, China, in 2005 and 2008, respectively, and the Ph.D. degree from Harbin Institute of Technology, Harbin, China, in 2014.

He is currently an Associate Professor of computer science, East China Jiaotong University, Nanchang. His research interests are in the areas of machine learning, deep learning, and clustering algorithms.



Xiaofei Yang (Member, IEEE) received the B.S. degree from Suihua University, Suihua, China, in 2011, and the M.S. and Ph.D. degrees from Harbin Institute of Technology, Harbin, China, in 2014 and 2019, respectively.

He was a Post-Doctoral Researcher with the Department of Computer and Information Science, University of Macau, Macau, China, from 2020 to 2023. He is currently with the School of Electronic and Communication Engineering, Guangzhou University, Guangzhou, China. His research interests are in the areas of semi-supervised learning, deep learning, remote sensing, transfer learning, and graph mining.



Jiangtao Peng (Senior Member, IEEE) received the B.S. and M.S. degrees from Hubei University, Wuhan, China, in 2005 and 2008, respectively, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2011.

He is currently a Professor with the Faculty of Mathematics and Statistics, Hubei University. His research interests include machine learning and hyperspectral image processing.



Yifang Ban (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees from Nanjing University, Nanjing, China, in 1984 and 1987, respectively, and the Ph.D. degree from the University of Waterloo, Waterloo, ON, Canada, in 1996.

Before joining KTH Royal Institute of Technology (KTH), Stockholm, Sweden, in 2004, she was a tenured Associate Professor at York University, Toronto, ON, Canada. She is currently the Chair Professor and the Director of the Division of Geoinformatics, KTH Royal Institute of Technology, Stockholm, and an Associate Director of the Digital Futures, Stockholm. Her research interests include Earth observation big data analytics, machine learning/deep learning, and their applications in mapping urban and land cover, monitoring urbanization, wildfires, and other environmental changes, and assessing environmental impact. She has published extensively on these topics.

Dr. Ban was the Co-Chair of the ICA Commission on Sensor-Driven Mapping and the Co-Lead of the Group on Earth Observations (GEO) Initiative "Global Urban Observation and Information" from 2012 to 2022. She has been an Associate Editor and Guest Editor for major remote sensing journals and an Invited Expert for EU and national grant application evaluations. Since 2016, she has been an Invited Expert of the UN-Habitat Technical Committee on Human Settlements Indicators for UN Sustainable Development Goals (SDGs).



Nan Jiang (Member, IEEE) received the Ph.D. degree from Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2008.

From 2013 to 2014, he was a Research Scholar with the Complex Networks and Security Research Laboratory, Virginia Tech, Blacksburg, VA, USA. He is currently a Professor with the Department of Internet of Things and the Director of the Intelligent Sensor Networks Laboratory, East China Jiaotong University, Nanchang, China. He has published more than 60 papers in international journals and conferences. His research interests include the Internet of Things, cyber–physical systems, and social networks.

Dr. Jiang is a member of ACM and a Senior Member of China Computer Federation (CCF). He was the Program Chair and a Committee Member of many international conferences, such as CSS 2019, IEEE EUC 2017, and ISICA 2015.

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